

Performance or Governance? Serious Game and Workshop as a Tool for Fostering Awareness of Responsible AI

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Abstract

The increasing prominence of AI ethics and governance has led to an increase of principles and guidelines. Consequently, AI developers and users require not only technical expertise but also a robust understanding of responsible AI and its governance. To effectively promote AI governance, it is crucial to facilitate knowledge acquisition and motivate stakeholders. Serious games have proven valuable tools for fostering an understanding of complex social issues. This study presents the design and implementation of a card game and accompanying workshop aimed at cultivating awareness of responsible AI. Within the game, participants assume the roles of members of a company's AI strategy team, tasked with balancing AI performance goals with ethical considerations while responding to dynamic events and managing resources. Our experimental results demonstrate the effectiveness of the workshop in raising awareness of responsible AI through active engagement with the card game.

CCS Concepts

• **Human-centered computing** → **Interaction design**; • **Social and professional topics** → **Computing / technology policy**.

Keywords

Responsible AI, AI governance, serious games

ACM Reference Format:

Hiroataka Kaji, Jun Kuribayashi, Hikaru Ohtani, and Arisa Ema. 2025. Performance or Governance? Serious Game and Workshop as a Tool for Fostering Awareness of Responsible AI. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3706599.3719951>

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719951>

1 Introduction

The increasing integration of artificial intelligence (AI) into society brings with it not only enhanced convenience but also potential risks, underscoring the growing importance of AI ethics and governance across research, development, and deployment. Numerous principles, guidelines, and laws addressing AI governance [6, 14, 16] have been established, accompanied by research exploring the concept of responsible AI and potential harms [3, 19]. Furthermore, tools such as case studies and impact assessments are being developed to support ethical AI development [7, 20]. Looking ahead to a future where AI becomes critical infrastructure, researchers have emphasized the need for resilience against potential crises, such as disruptions in AI availability during disasters [11]. For AI developers and users, understanding and implementing AI governance guidelines and tools are essential. This necessitates not only technical proficiency in algorithms but also a broad range of expertise encompassing ethics, law, and practical experience. However, opportunities to learn from specific incidents or crises, such as deploying AI in disaster response, are limited. Therefore, facilitating accessible knowledge acquisition and fostering motivation is crucial for advancing responsible AI and its governance.

Serious games and role-playing have emerged as effective methods for enhancing comprehension of complex social issues. Numerous games addressing diverse challenges, including cybersecurity [1, 10, 21], health and wellness [4, 22], pandemic [12], and environmental concerns [13, 18], have demonstrated their efficacy. These games utilize various formats, including card games and online platforms. While the intersection of AI and serious games often focuses on AI support for game development [17], several studies have explored the use of serious games for promoting responsible AI. Avin et al. highlighted the value of role-playing in examining the future of AI and its societal impact, arguing that it effectively clarifies both positive and negative consequences for individuals and organizations [2]. Hanschke et al. investigated the long-term impact of role-playing exercises designed for ethical emergency response in data science [8]. Hardebolle et al. employed online games within workshops to engage students with ethical dilemmas [9].

This study introduces a serious game and accompanying workshop designed to support beginner to intermediate levels of responsible AI and AI governance knowledge. Our card game casts

participants as members of a company's AI strategy team, challenging them to balance AI performance with ethical considerations to optimize operational efficiency and cultivate trust. Through collaborative discussion and resource allocation, teams navigate events and accidents, accumulating "trust points." This design fosters interaction and deliberation on AI governance, encouraging teamwork to maximize trust points while simultaneously pursuing individual secret missions. The game dynamically illustrates the vulnerabilities that can arise from prioritizing performance over governance and highlights the interconnectedness of stakeholders and AI-related incidents. To evaluate its effectiveness, we conducted a workshop incorporating the game and analyzed gameplay data and questionnaire responses to demonstrate its impact on raising awareness of responsible AI and governance.

2 Serious Game for Responsible AI

2.1 Goal and Approach

Understanding and practicing responsible AI and AI governance can be envisioned as a multistage journey, progressing from novice to expert levels. When designing serious games or role-playing exercises, the required preliminary knowledge varies depending on whether the objective is to provide a broad overview of responsible AI or to delve into specific risk events and concrete scenarios. This card game is designed for individuals with a beginner to intermediate understanding of responsible AI, including company employees, local government officials, and students. By offering an engaging and accessible experience, the game aims to introduce key concepts in responsible AI and motivate participants to further their learning in this domain. The proposed card game presents a stylized version of real-world scenarios, where the consequences of players' decisions are amplified to highlight their impact.

Drawing inspiration from existing serious games [5, 15], we incorporated the following design principles:

- (1) The game mechanics incorporate dilemmas arising from resource constraints, encouraging negotiation and discussion through a combination of team collaboration and individual competition.
- (2) When played with multiple teams, trust points are publicly visible, fostering competition and strategic gameplay.
- (3) The game features a simple, expandable structure with elements of randomness, allowing for varied gameplay and strategic adaptation across multiple sessions.
- (4) The game aims to raise awareness of responsible AI by exposing players to the diverse stakeholders involved in AI systems and demonstrating the potential consequences of prioritizing AI performance over ethical considerations and organizational resilience.

2.2 Game Design

The card game, depicted in Figure 1, comprises six key elements: a game board, resource cards, event cards, accident cards, role cards, and secret mission cards.

Rules: Players form teams of 3–6 individuals, each assuming the role of a department within a major corporation or startup. Through collaborative discussion, the team strategically allocates computing and human resources on the game board with the aim of enhancing AI performance. Teams accumulate "trust points" based on resource allocation and AI utilization events, while accidents can

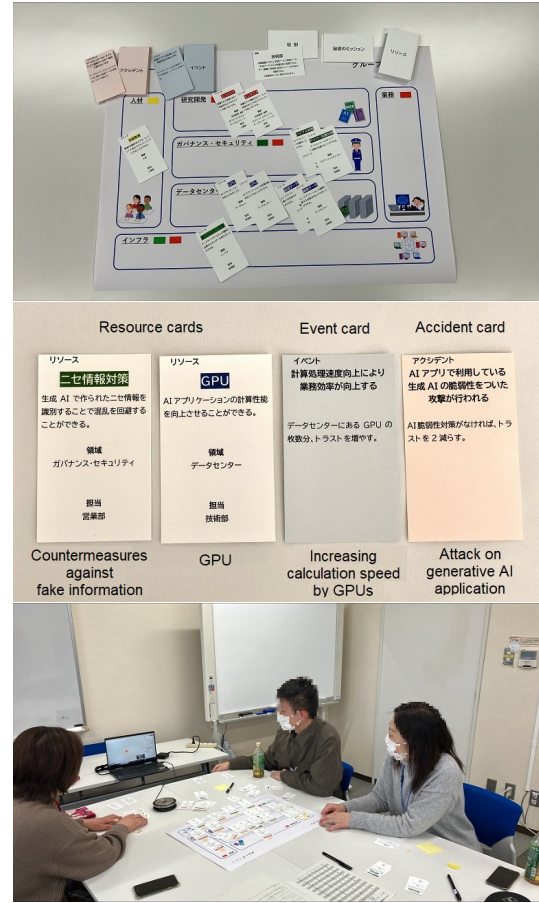


Figure 1: The developed card game (upper: overview middle: cards) and a gameplay scene (lower). The game board and cards are in Japanese to accommodate the participants' native language.

lead to a loss of trust points. The game unfolds over multiple turns, each involving three distinct steps: event, accident, and resource allocation. The primary objective is to maximize trust points as a team. Simultaneously, each player pursues a secret mission assigned to them through negotiations and strategic maneuvering.

Game board: The game board is structured around six key areas related to AI: Research and Development, Governance and Security, Datacenter, Infrastructure, Human Resources, and Operations. Resource cards are strategically placed within these corresponding areas. For example, a "GPU" card can only be allocated to the Datacenter, while an "Engineer" card can be assigned to Research and Development, Governance and Security, Datacenter, Infrastructure, or Operations.

Resource cards: These cards represent resources essential for AI development and operation. Teams strategically allocate these resources to enhance AI performance and bolster resilience against potential accidents. Resource cards are distributed to teams in advance and encompass over 50 cards across 24 distinct types, broadly categorized as follows:

Human resources: This category includes engineers, outstanding engineers, AI ethics experts, and AI safety experts. Deploying engineers in specific areas improves performance and mitigates trust point losses from accidents. Outstanding engineers and AI ethics experts represent valuable human resources that can only be acquired through training existing engineers or external recruitment (achieved by placing a card in the Human Resources area to exchange an engineer for a specialized resource).

Physical and data resources: These cards represent the foundational elements of AI performance, including data (images, language, etc.), GPUs, and infrastructure enhancements. Accumulating data and GPUs generates trust points during events by, for instance, improving operational efficiency. Infrastructure enhancement cards mitigate the impact of accidents and help preserve trust points.

Countermeasures: This category encompasses governance elements such as transparency and fairness, and security measures, such as countermeasures against fake information [14]. These cards mitigate the impact of accidents.

The limited capacity for resource placement on the game board creates strategic dilemmas and necessitates careful resource management.

Event cards: These cards depict positive events stemming from AI introduction and utilization, such as resolving company challenges, optimizing operational efficiency, and enhancing customer satisfaction. When resource cards are present in areas corresponding to the event card, the team gains the indicated number of trust points. For example, an event card representing improved operational efficiency owing to increased AI calculation speed would award trust points equal to the number of "GPU" cards allocated to the relevant area.

Accident cards: These cards represent incidents arising from AI systems, crises triggered by AI–human interactions, or unforeseen disasters. The team incurs a loss of trust points, as indicated on the card. Accident scenarios were carefully selected based on various guidelines and research papers to reflect potential real-world occurrences [14, 19]. For example, an attack on a generative AI application might result in a loss of two trust points if the "AI Vulnerability Countermeasures" card is not placed. While most accidents are temporary, causing a one-time loss of trust points, certain events, such as natural disasters, can have a prolonged impact, leading to consecutive trust point deductions over several turns. To mitigate such risks, teams must strategically allocate resources to strengthen infrastructure and develop human resources, even if these resources do not directly contribute to immediate AI performance gains.

Event and accident cards are placed face down, and one card is drawn at the beginning of each turn, introducing an element of uncertainty and requiring adaptive strategies.

Role cards: Based on the number of participants, roles such as technical department, sales department, and management team are assigned before the game commences. Each role has defined responsibilities and authority, influencing resource prioritization during initial allocation and final decision-making in each turn. For instance, the technical department has the authority to prioritize two GPUs or data cards during the initial resource allocation phase.

Secret mission cards: These cards outline individual missions that players must strive to achieve, such as prioritizing performance

or environmental protection. These missions remain confidential, adding a layer of complexity and intrigue to the game. For example, a secret mission might require a player to ensure that four or more GPU and data cards are in place by the end of the game.

The interplay of roles and secret missions encourages negotiation and strategic maneuvering. Players may leverage the authority of their role to allocate specific resources while subtly hindering the allocation of others to fulfill their secret objectives.

2.3 Procedure

(1) **Role assignment:** Players determine their roles and secret missions. Roles can be assigned randomly or through team discussion and are then publicly revealed. Secret missions are randomly assigned and remain private to each player.

(2) **Initial resource allocation:** Teams engage in discussion to allocate their initial 10 resource cards across the 6 AI component areas on the game board. All resources are visible to all players. During this initial allocation, certain roles have the authority to prioritize specific resources. However, exercising this authority is optional. The allocation of remaining resources is determined through team discussion, requiring permission from the role responsible for that specific resource type.

(3) **n th turn (Event and Accident):** The team or moderator draws an event card and an accident card. The team gains trust points based on the event's content and loses trust points based on the accident's content, based on the resources allocated to relevant areas. The drawn event and accident cards are then discarded.

(4) **n th turn (additional allocation or reallocation of resources):** The team selects one unplaced resource card and adds it to the game board. The maximum number of resource cards allowed on the board is 20. If this limit is reached, the team can either reallocate an existing "Engineer" card or exchange an allocated resource card with an unplaced resource card. Resource allocation requires permission from the role responsible for that resource type. The n th turn ends, and the game returns to step (3). The game concludes after 20 turns.

To facilitate understanding of the game mechanics and the impact of initial resource allocation, only event cards are drawn for the first five turns. Accident cards are introduced from the sixth turn onwards, increasing the complexity and requiring adaptive strategies. In the middle stage of the game, discussions focus on the strategic addition of new resources in response to fluctuations in trust points. In the latter half, as the resource limit is reached, teams must prioritize resource reallocation while simultaneously pursuing their secret missions.

2.4 Parameter Tuning

Given the game's simple rule of adjusting trust points based on resource card allocation, careful parameter tuning is essential. Key parameters include the number of resources and the specific calculations for trust point adjustments. The element of randomness introduced through card effects also plays a significant role. To analyze these parameters and optimize game balance, we simulated resource allocation and evaluated trust accumulation over multiple turns.

We focused on two primary strategies: a performance-oriented strategy that prioritizes trust gain through events (primarily by

allocating GPUs and data) and a governance-oriented strategy that aims to minimize trust point loss from accidents (by focusing on human resource development and incident countermeasures). For simplicity, the initial allocation was fixed at 15 resources without any additions or reallocations. The objective of this analysis was to achieve a balance in final trust points between the two strategies. While the performance strategy facilitates rapid trust accumulation, it increases vulnerability to accidents. Conversely, the governance strategy offers greater resilience against accidents but results in slower trust gain. Figure 2 illustrates the trust point transitions observed using the adjusted parameters, showcasing evaluations of both strategies across three random event/accident sequences. Although the average trust points after 20 turns were similar for both strategies, the performance strategy exhibited greater variability. In actual gameplay, a hybrid strategy incorporating elements of both performance and governance strategies is likely to be adopted.

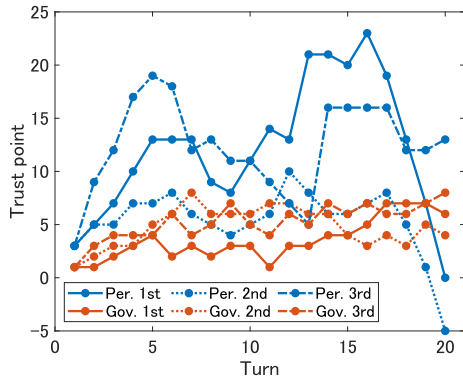


Figure 2: Simulation results. Blue lines represent the performance-oriented strategy, whereas red lines represent the governance-oriented strategy. Solid, dotted, and dashed lines correspond to the first, second, and third simulations, respectively.

2.5 Workshop using Serious Game

This workshop aimed to address the following research question:

Research question: Can the game and workshop increase awareness of AI governance among participants with beginner to intermediate levels of knowledge?

The workshop employed a two-phase approach. Participants first engaged in a round of the card game with minimal prior knowledge of its content. This was followed by a structured reflection session, after which participants played the game a second time. During the reflection session, guided by a worksheet, participants analyzed the outcomes of the first game, focusing on the decision-making processes that led to those results. They then engaged in team-based discussions to identify desirable outcomes and strategies to achieve them. The second round of gameplay reinforced learning by providing an opportunity to apply insights from the reflection session, emphasizing strategies for maximizing trust points, the importance of balancing performance with governance to mitigate risks, and key concepts in AI ethics and governance. Multiple teams

participated in the workshop simultaneously. A moderator drew event and accident cards, ensuring that all teams experienced the same sequence of events. Furthermore, each team's trust point transitions were displayed in real time. This design choice reflects the dynamics of real-world AI services, where internal governance practices may be opaque, but the reputation of services is readily disseminated through social media and other channels.

3 Experiment

To evaluate the effectiveness of the game and workshop, we conducted an experiment approved by our institution's ethics committee. The participants were given a prior explanation, and consent was obtained. The participants comprised 13 individuals from the research department of a company: nine Japanese engineers and four administrative staff members (six female, seven male). Their ages ranged from their 20s to their 60s, with the majority in their 40s. According to a pre-workshop questionnaire, approximately 80% of the participants possessed beginner to intermediate levels of knowledge regarding AI technology and AI governance. Eleven participants attended the workshop in person, while two participated remotely. Participants were divided into three teams (A, B, and C), each supported by a facilitator. A designated moderator oversaw the game, drawing and announcing events and accident cards.

Pre-workshop questionnaire: This questionnaire assessed participants' knowledge of AI, frequency of AI use, and familiarity with AI governance concepts. To gauge potential shifts in awareness of AI governance before and after the workshop, participants were asked to select 10 out of 20 items they considered important in AI services. Eight of these items pertained to AI governance, encompassing aspects such as security and fairness. The order of the 20 items was randomized for each participant and for both the pre- and post-workshop questionnaires.

Workshop: The workshop commenced with a 10-minute introduction outlining the objectives and activities. Participants then engaged in the first round of the card game (30–40 min). Following this, each team used a worksheet to reflect on the game outcomes, drawing connections between the game dynamics and real-world human–AI interactions. Teams documented their key takeaways from the first game, including decisions made and accidents encountered, using sticky notes. They then engaged in a 15-minute discussion about desirable future scenarios and the actions needed to achieve them. After a 10-minute break, participants played the second round of the game (30–40 min), followed by a debriefing session to consolidate learning.

Post-workshop questionnaire: This questionnaire mirrored the pre-workshop questionnaire, assessing participants' awareness of AI governance. Additionally, participants rated the game's entertainment and educational value on a five-point scale. Open-ended questions elicited feedback on key discussion points, memorable episodes, reflections on the learning experience, and suggestions for game improvement.

4 Result

4.1 Game's Progress

Figure 3 depicts the transitions in trust points for each team across both rounds of the game. It is important to note that the sequence of events and accidents differed between the first and second rounds. In the first round, a major accident (earthquake) occurred during the 15th turn, simulating a situation where AI systems became unusable for three consecutive turns. Although assigning restoration personnel could mitigate the impact on the duration of this disruption, teams A and C experienced a significant drop in trust points owing to this event. Conversely, team B, having adopted a governance-oriented strategy from the outset, was less affected and secured first place with 15 points.

Informed by their reflections on the first round, each team adopted a distinct strategy in the second round. Team A implemented a balanced approach in their initial resource deployment. In the latter stages of the game, they proactively reduced their reliance on GPUs to minimize potential point losses and shifted resources toward governance measures and human resource development. This strategic adaptation resulted in a first-place finish with 22 points. Team B maintained a governance-oriented strategy similar to their approach in the first round, making minimal changes to their resource allocation even after the 11th turn, when reallocation became possible. Consequently, they achieved the same score of 15 points as in the first round, securing third place. Team C adopted a performance-oriented strategy, prioritizing GPUs and data in their initial deployment and accumulating points rapidly until the sixth turn when the first accident occurred. However, in the second half of the game, the gains from events were largely offset by losses from accidents, hindering their progress. Notably, team C demonstrated preparedness for potential earthquake disaster accidents by maintaining readily available restoration personnel throughout the first round. The results of teams B and C align with the simulation findings. Team A's success highlights the effectiveness of dynamic strategy adaptation in response to evolving game conditions.

4.2 Questionnaire

Awareness of AI governance: Out of the 130 items selected by the 13 participants, the total number of AI governance-related items chosen increased from 57 (43.8%) before the workshop to 62 (47.7%) after. While eight participants individually selected more AI governance items after the workshop, three selected fewer. However, this difference was not statistically significant ($p < 0.05$) in the Wilcoxon signed-rank test. Notably, participants predominantly selected items related to privacy and security, whereas those related to resilience, such as environmental concerns and AI dependency, were rarely chosen. This trend remained consistent before and after the workshop.

Impressions of the game: Nine out of 13 participants reported finding the game rules challenging to grasp initially. This difficulty may stem from the volume of information presented at the beginning of the game, encompassing resource types, event and accident card content, and trust calculations. Despite this initial challenge, over nine participants expressed positive views regarding the game's entertainment and educational value (scoring 4 or higher on a 5-point scale), and nine participants found the playing

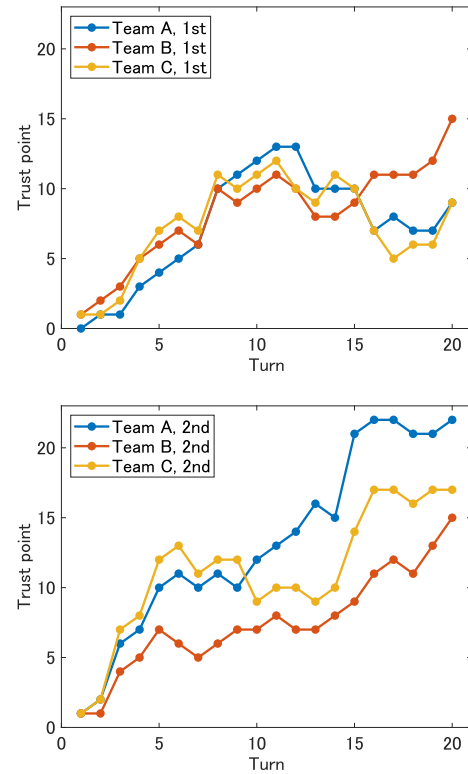


Figure 3: Experiment result of first (upper) and second rounds (lower). Blue, red, and yellow lines indicate teams A, B, and C, respectively.

time appropriate. These responses suggest that the game achieved a satisfactory level of overall quality. Regarding the level of interaction, eight participants reported active discussions (scoring four or higher on a 5-point scale), whereas three rated the discussions as moderate (3 on a 5-point scale), indicating successful facilitation of interactive engagement.

Noteworthy episodes: The shared experience of responding to the same events and accidents, coupled with the real-time display of trust points, fostered a sense of competition among teams, positively influencing team discussions and strategy reviews. Some participants remarked on the game's realism, drawing parallels with real-world scenarios. For instance, they noted how the initial focus on performance and trust point accumulation through AI research and development in the first half of the game mirrored the early stages of AI adoption. The shift toward maintenance and human resource development in the second half, in anticipation of potential accidents, reflected the growing emphasis on long-term sustainability and risk mitigation in AI deployment.

Ideas for improving the game: Several participants offered suggestions for enhancing the clarity of the rules. To address the challenge of remembering the diverse range of events and accidents, participants proposed displaying them on a screen for easy reference. Regarding roles and secret missions, observations revealed a divergence in team engagement, with some teams actively

negotiating while others remained largely unaware of these mechanics. While these elements can enhance interaction, they also contribute to the complexity of the rules. Therefore, delaying their introduction until subsequent rounds of play may be beneficial.

4.3 Discussion

Analysis of the game progress and questionnaire responses suggests that the proposed game and workshop can effectively enhance awareness of responsible AI and AI governance. While acknowledging the potential influence of varying event and accident sequences, the observed trust point transitions demonstrate that participants steadily accumulated trust points in the first half of the second game and experienced fewer point deductions in the second half compared to the first. Furthermore, the reflection sheets and questionnaire responses reveal that many participants recognized the importance of governance measures and human resource development in mitigating the impact of potential accidents. This heightened awareness was reflected in their strategic decision-making during the second round of gameplay. Despite these promising findings, this study has limitations. The primary limitation is the small sample size and the homogeneity of the participant pool. Although the participants possessed a certain degree of expertise diversity, they were all employees of the same company. Additionally, further research is needed to investigate the long-term impact of the game and workshop on behavior related to AI governance.

5 Conclusion

This study presented the design and implementation of a card game and accompanying workshop aimed at raising awareness of the importance of responsible AI and AI governance. Through an experimental workshop, we demonstrated the potential of this game-based approach to cultivate interest in responsible AI among participants with beginner to intermediate levels of knowledge. In the future, we plan to refine the game rules and parameters based on the insights gained from this initial experiment and revise the workshop design accordingly. This will enable us to conduct workshops with diverse participants from various fields and levels of expertise. In particular, we recognize the need to tailor the game and workshop to different organizational contexts, such as companies and local governments, which may have varying perspectives on AI implementation and human resource allocation. We intend to systematically analyze these similarities and differences to inform revisions to the game content and ensure its adaptability across diverse settings.

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